MS AI Project Team 13

3D PRINT ERROR SOLUTION

NOpaghetti

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3D Printer Industry Status & Outlook

Market Size

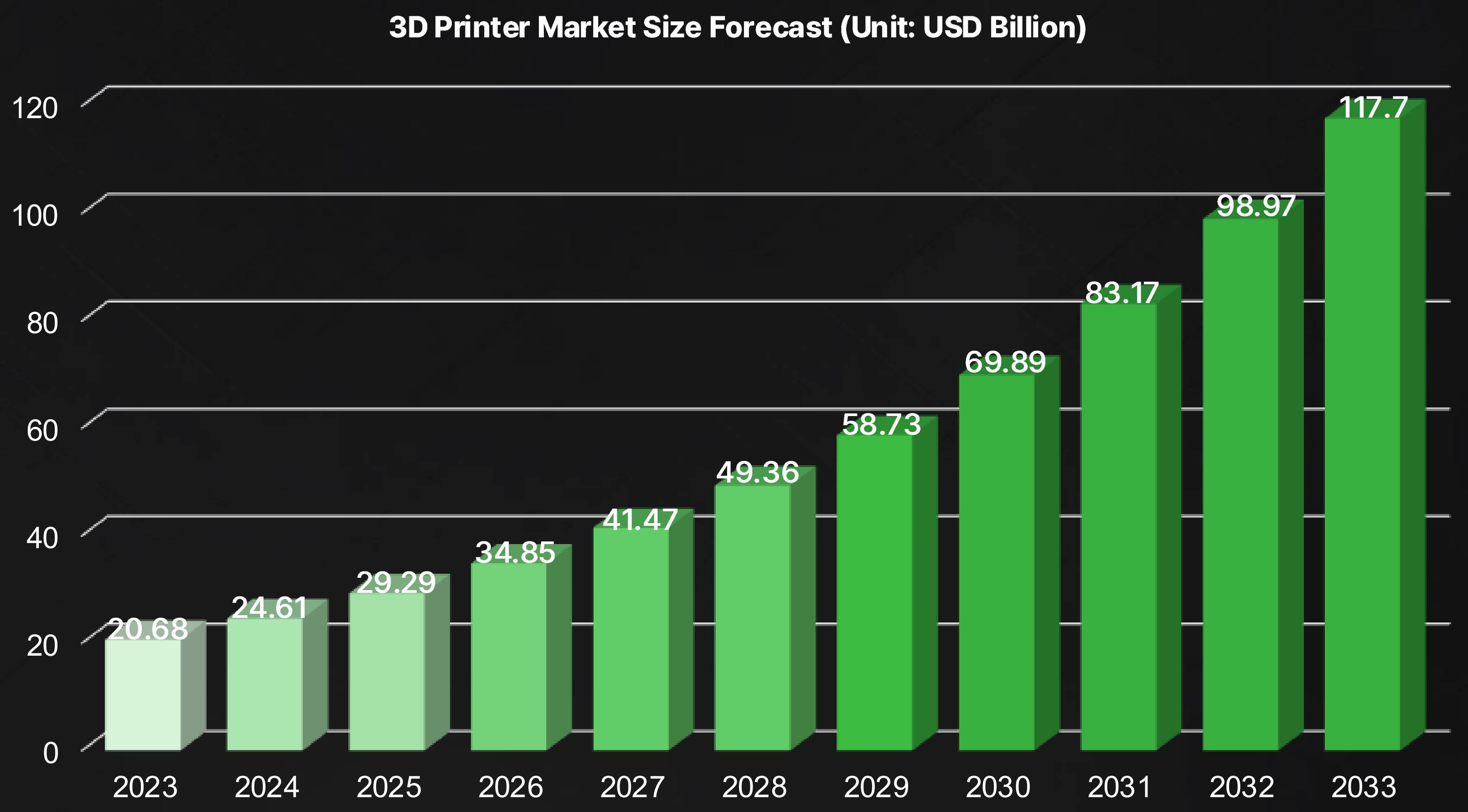
$21.1B in 2023
2032 $118.9B USD Projected

Industry Status

Falling costs fuel parts & prototyping adoption

Rapid Growth Forecast

2024–2032
~19% CAGR Expected (Exceptionally High)



OVERVIEW

Reason for Topic Selection

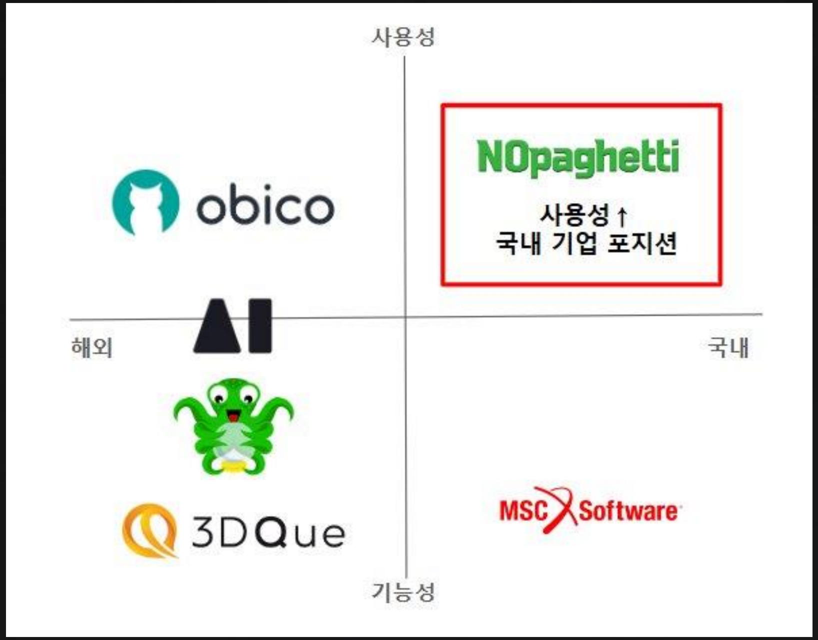
- 2021 HSSMI (UK Mfg. Consultancy) Case Study: Reports 3D Printing Outputs Discarded at 33% (~379,000kg waste per 1,148,400kg consumed)
- 80% of 3D printing waste is caused by print failures
- 3D print errors waste significant time and electricity Early error detection improves UX & reduces carbon emissions

We selected this topic to prevent resource waste by catching 3D print errors, enabling eco-friendly manufacturing and contributing to society through this project.

OVERVIEW

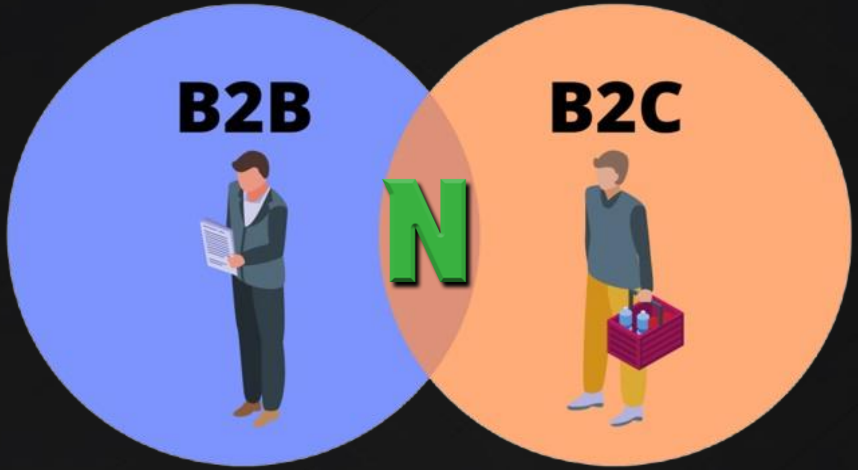
Market Strategy

Domestic Market First-Mover



- Overseas programs → **Korean users face usability challenges**
- Korean-language customer support, community driven info sharing & feedback
- New market entry strengthens **market position**

Via B2B & B2C: Diverse Customer Base Secured



Manufacturers
Prototyping Companies
R&D & Labs etc.

Annual License Sales

Individual Users
Independent Creators etc.

Subscription Sales

Dual-targeting strategy enhances brand image

Team Members & Roles

Junsang Lee
Project Manager

Duhyeon Nam
Frontend Engineer

Jooyoung Hur
Backend & Server Engineer

Yujin Lee
Data & ML Engineer

Wonhyeong Son
Market & Data Analyst

Taehwan Choi
Presentation Designer

TIMELINE

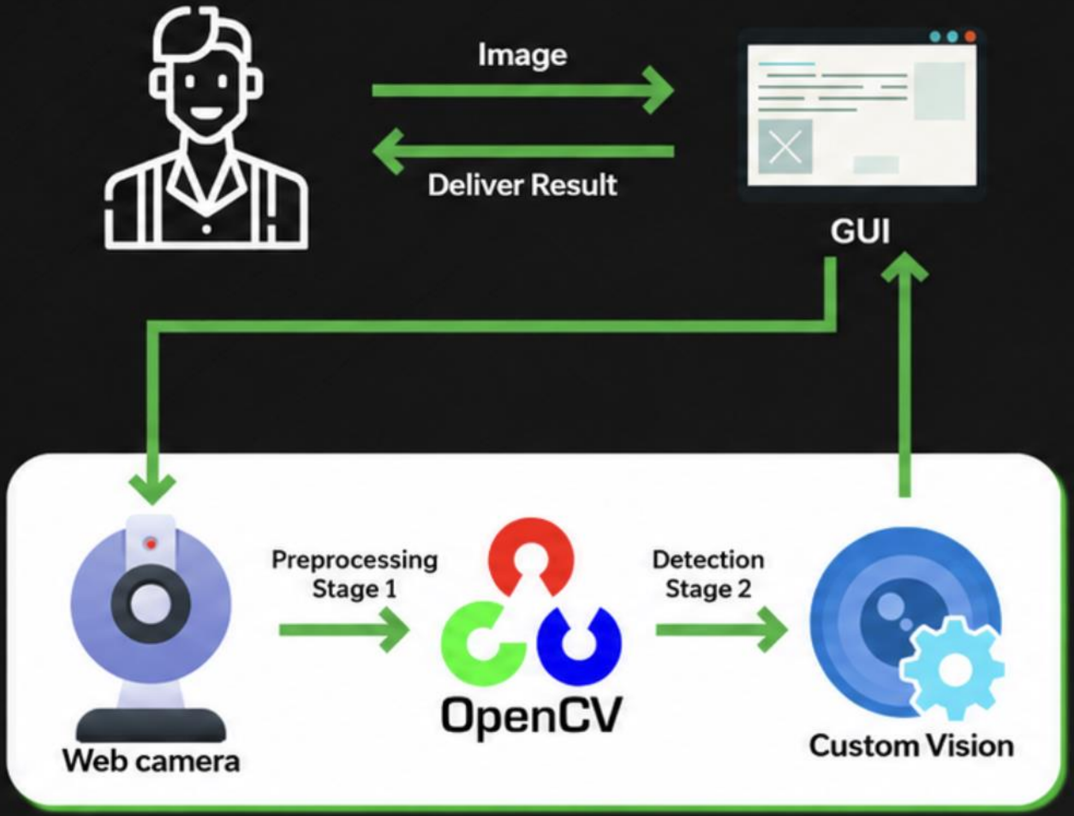
Project Timeline

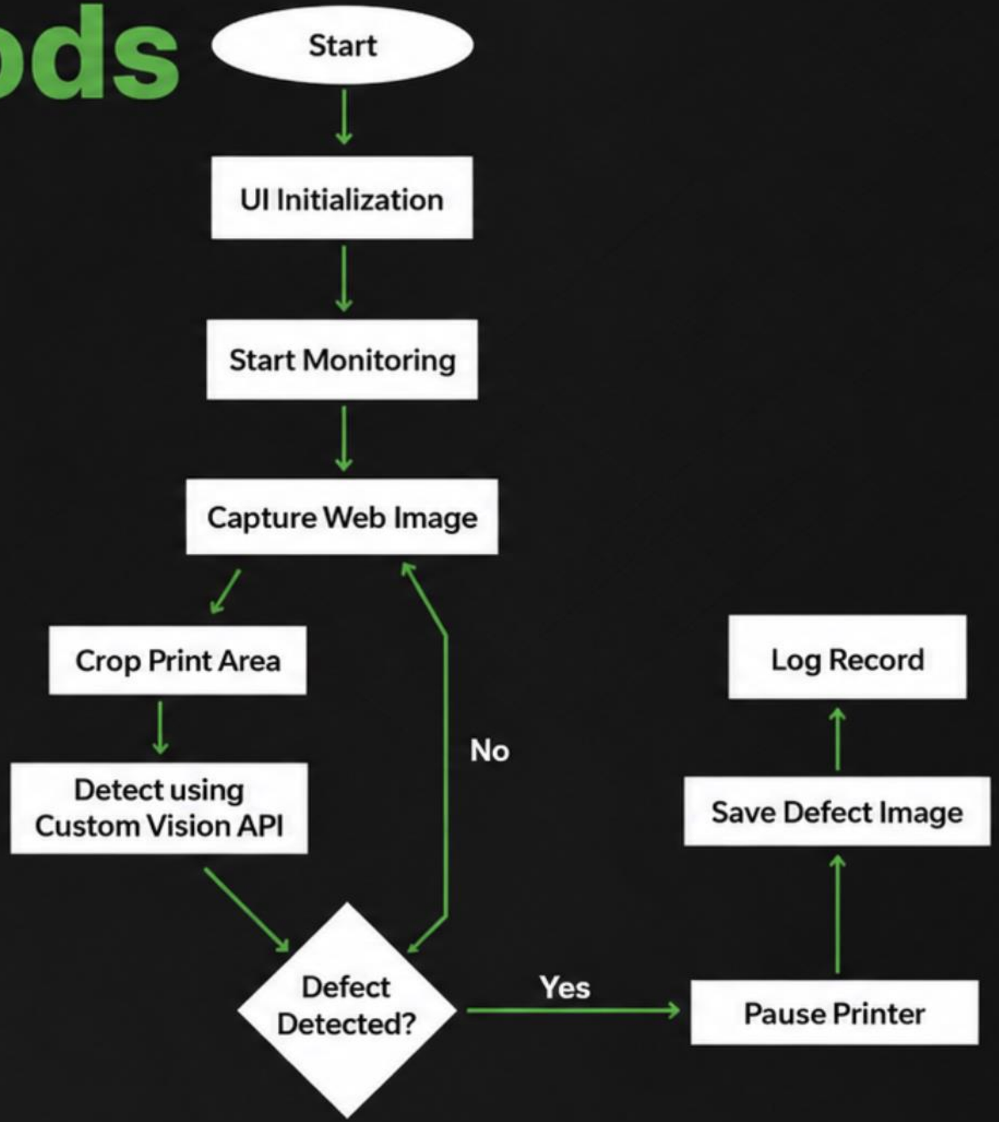
10/28 (Mon)	10/29 (Tue)	10/30 (Wed)	10/31 (Thu)	11/1 (Fri)
			Topic Selection	Initial Planning Data Collection

11/4 (Mon)	11/5 (Tue)	11/6 (Wed)	11/7 (Thu)	11/8 (Fri)
Custom Vision Tagging/Classification Research Summary	PPT Design & Writing App UI Implementation	Defect Detection Implementation Control Feature Implementation	Code Team Review PPT Final Review Code Refactoring	Presentation Rehearsal

PROGRESS

Procedure & Methods





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PROGRESS

Procedure & Methods

Tech Stack & Platforms

Custom Vision

API& Endpoint Integration
Real-time Image Analysis
Defect Location Visualization

Front End

Native application
-level performance required

Azure Custom Vision
PyQT
MS Azure
Kaggle
Python
OpenCV
Roboflow

Data Preprocessing

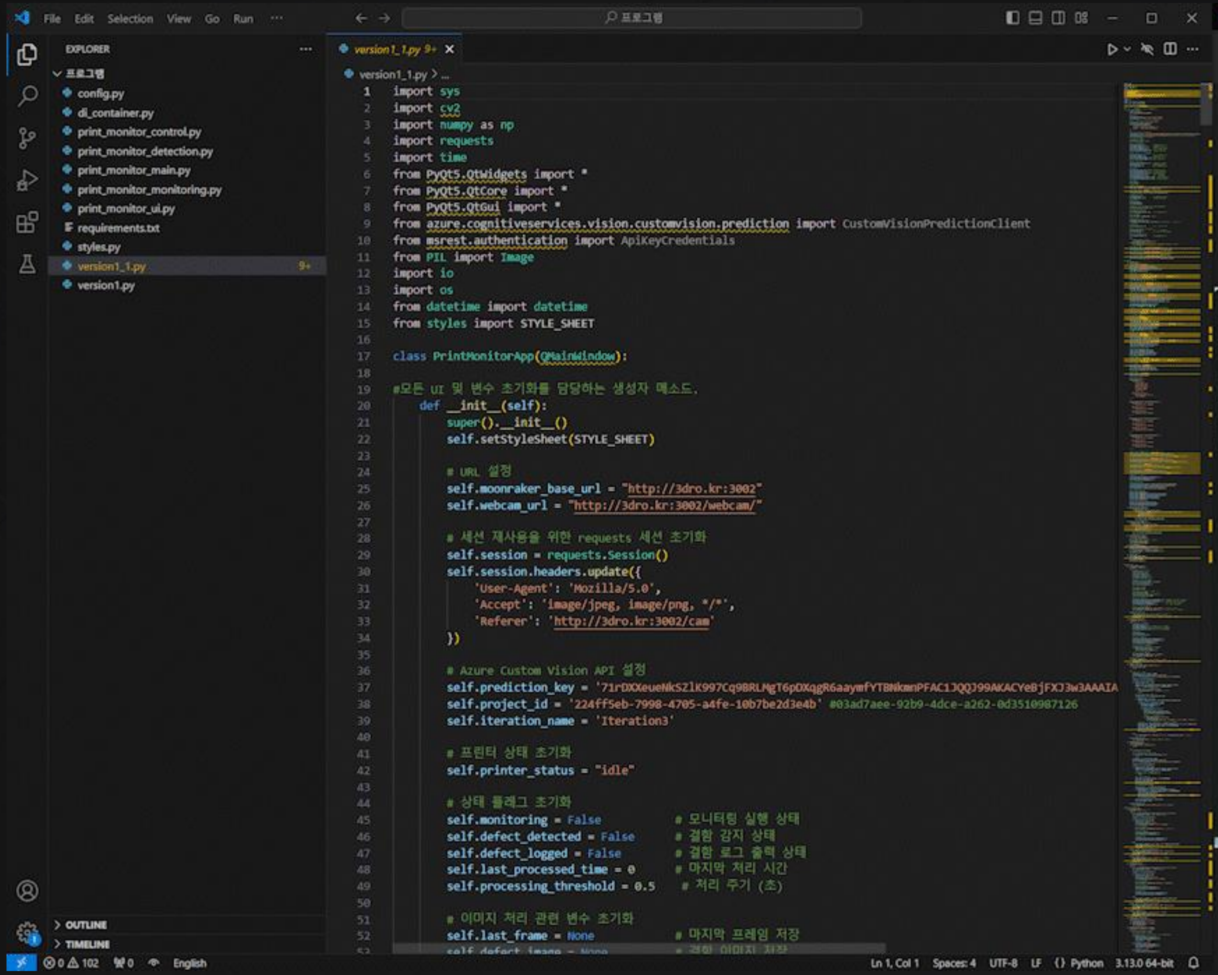
Defect Detection Threshold
False Positive Minimization

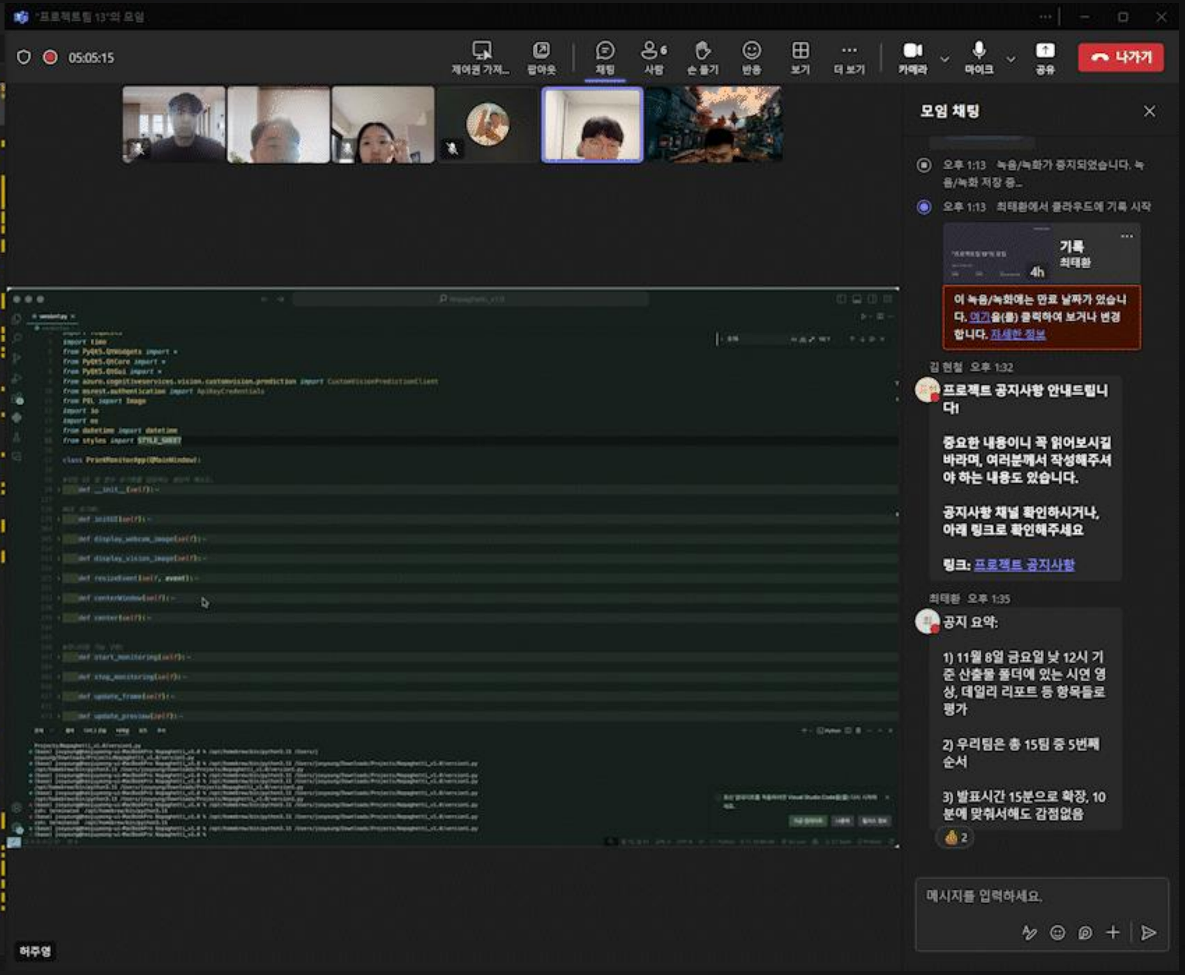
Back End

Library Utilization
OpenCV / Numpy
PIL / Requests

PROGRESS

Code Team Review



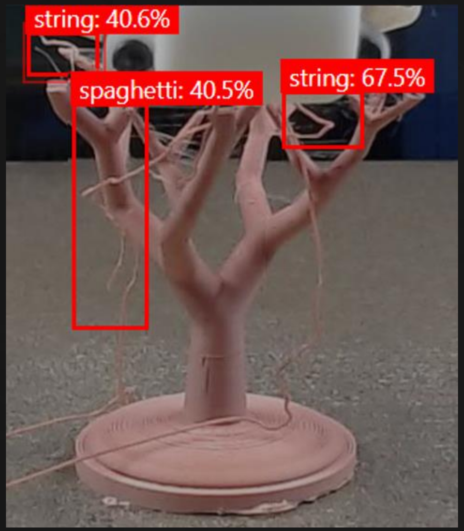


PERFORMANCE

Results

Initial Training	Revised Training Strategy	Outcome
Trained with maximum images 3DNozzle / floor dust falsely detected — frequent false positives	Reduced dataset size; trained on clearly distinctive images only	Significant error reduction (Quality over quantity matters)

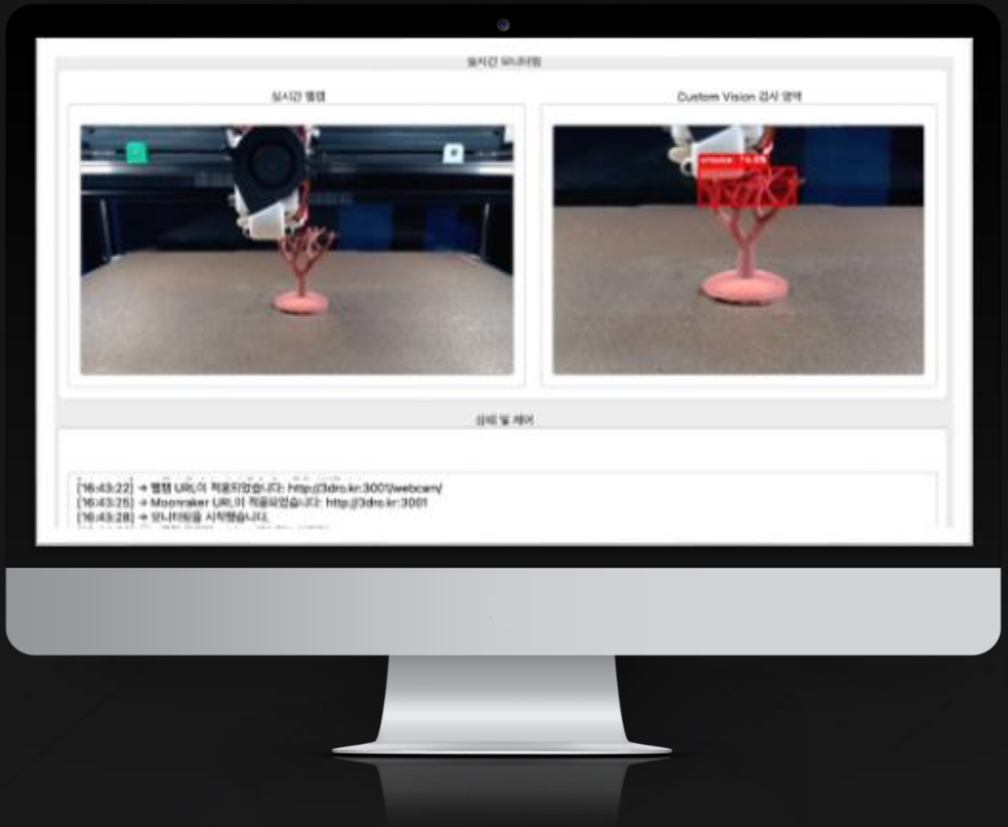
Final Test Result Images



PERFORMANCE

Results

Demo Video



모니터링 설정

Webcam URL: http://3dro.kr:3001/webcam/ 적용

Moonraker URL: http://3dro.kr:3001 적용

웹캠 및 서버 링크 입력

실시간 모니터링

실시간 웹캠

Custom Vision 검사 영역

상태 및 제어

[16:59:56] → 프린터 상태가 '대기 중'(으)로 변경되었습니다

▶ 모니터링 시작 ■ 모니터링 중지 ❚❚ 일시정지 ▶ 재개

AI ETHICS

AI Ethics Application

Privacy Protection & Security

✓ Terms of service consent provided

✓ **Azure Cloud security services for data storage**

Reliability

✓ **Accurately identifying print errors through continuous error detection algorithm improvement**

Accountability

✓ Errors are logged and traceable

✓ Customer support system activated for error handling

Fairness

✓ **Continuous data training and updates for diverse environment compatibility**

Inclusivity

✓ Program guides and support services provided

✓ Easy to use — intuitive UI

Transparency

✓ **Program's error identification criteria and notification standards — clearly communicated to users**

THANK YOU

Thank you!

Q & A